After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the three subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

3.1 · Alcohols: names, classes, properties

Q1 CH3-LO1

Classify each alcohol.

Options: 1° · 2° · 3°

$CH_3CH_2CH_2OH$ _____

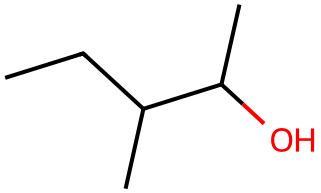
$CH_3CH(OH)CH_3$ _____

$(CH_3)_3COH$ _____

cyclohexanol _____

Q2 CH3-LO1

Name this alcohol.



A. 3-methyl-2-pentanol

B. 3-methyl-4-pentanol

C. 2-ethyl-1-butanol

D. hexanol

Q3 CH3-LO1

Arrange in order of increasing water solubility: 1-butanol, methanol, 1-octanol. (Ball 14.E)

A. 1-octanol < 1-butanol < methanol

B. methanol < 1-butanol < 1-octanol

C. 1-butanol < methanol < 1-octanol

D. all equal

Q4 CH3-LO1

Why is propylene glycol, but not ethylene glycol, used as a drug solvent?

A. Propylene glycol is metabolised to harmless pyruvate; ethylene glycol to toxic oxalate

B. Ethylene glycol has no –OH

C. Propylene glycol is insoluble in water

D. They are the same compound

3.1 · SN1 and SN2

Q5 CH3-LO2

Match each feature to SN1 or SN2.

Options: SN1 · SN2

one step _____

goes through a carbocation _____

best for 1° haloalkanes _____

best for 3° haloalkanes _____

rate depends on [R–X] only _____

inverts a chiral carbon _____

Q6 CH3-LO2

Which haloalkane reacts fastest by SN2?

A. CH_3CH_2Br

B. $(CH_3)_3CBr$

C. $(CH_3)_2CHBr$

D. they are equal

Q7 CH3-LO2

Which is the best leaving group?

A. I^-

B. Cl^-

C. F^-

D. OH^-

Q8 CH3-LO2

Name the alcohol formed when $(CH_3)_3CBr$ is warmed with water.

_____

Q9 CH3-LO2

Bromoethane is heated with KOH dissolved in ethanol. The main organic product is:

A. ethene
B. ethanol
C. diethyl ether
D. ethanal

3.1 · Dehydration and oxidation

Q10 CH3-LO3

Give the product of each alcohol with acidified $K_2Cr_2O_7$.

Options: 2-pentanone · cyclohexanone · no reaction · ethanoic acid · ethanal

2-pentanol _____

cyclohexanol _____

2-methyl-2-propanol _____

ethanol (excess oxidant) _____

Q11 CH3-LO3

2-Methyl-2-propanol with excess conc. H_2SO_4 at 180 °C gives: (Ball 14.E)

A. 2-methylpropene (isobutylene)
B. 2-methylpropanal
C. di-tert-butyl ether
D. no reaction

Q12 CH3-LO3

$C_4H_{10}O \rightarrow C_4H_8O$ is an oxidation. (Ball 14.E)

True / False

Q13 CH3-LO3

Why is methanol far more dangerous to drink than ethanol?

A. It is oxidised to formaldehyde and formic acid
B. It is a thiol
C. It cannot be metabolised
D. It is an alkane

3.2 · Ethers and thiols

Q14 CH3-LO4

Name $CH_3OCH_2CH_2CH_3$. (Ball 14.E)

A. methyl propyl ether
B. butanol
C. propyl methanol
D. methoxyethane

Q15 CH3-LO4

Diethyl ether (bp 35 °C) and 1-butanol (bp 117 °C) are isomers. Which statement is true?

A. Both dissolve in water to a similar small extent (about 8 g/100 mL)
B. The ether hydrogen bonds to itself
C. 1-Butanol is insoluble in water
D. The ether boils higher

Q16 CH3-LO4

Write the condensed formula of tert-butyl mercaptan, a natural-gas odorant. (FGOB 14.8)

_____

Q17 CH3-LO4

Mild oxidation of two thiol molecules gives:

A. a disulfide R–S–S–R

B. an ether

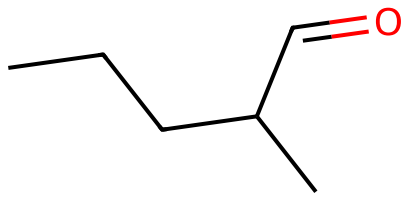
C. a sulfate

D. an alkene

3.3 · Aldehydes and ketones

Q18 CH3-LO5

Name this compound.



A. 2-methylpentanal

B. 4-methylpentanal

C. 2-methylpentanone

D. hexanal

Q19 CH3-LO5

Which has the highest boiling point? (similar molar masses)

A. propanal

B. butane

C. methoxyethane

D. 1-propanol

Q20 CH3-LO5

Predict each test result for propanal and propanone.

Options: silver mirror · brick-red precipitate · no change

propanal + Tollens' _____

propanone + Tollens' _____

propanal + Benedict's _____

propanone + Benedict's _____

Q21 CH3-LO5

Reduction of butanone with H_2/Ni gives:

A. 2-butanol
B. 1-butanol
C. butanal
D. butanoic acid

Pharmacy scenario · The diabetic patient

Scenario

Encik Faizal has type 1 diabetes and has missed his insulin for two days. His wife notices a fruity, nail-polish-remover smell on his breath. At the pharmacy counter he asks for something for nausea. You also know that old urine-sugar tests used **Benedict's reagent**.

Q22 CH3-LO5

Which compound causes the smell on his breath?

A. acetone (a ketone)
B. ethanol
C. diethyl ether
D. ethanethiol

Q23 CH3-LO5

Benedict's reagent detected glucose in urine because glucose behaves like:

A. an aldehyde (a reducing sugar)
B. a ketone that resists oxidation
C. an ether
D. a thiol

Q24 CH3-LO5

What should the pharmacist do, and why?

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 14 and exercises 14.E, and *Fundamentals of General Organic and Biological Chemistry* 14.4–14.9 and 15.1–15.5, LibreTexts, CC BY-NC-SA 3.0. SN1 and SN2 items, colour tests and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Organic Compounds that Contain Oxygen and Sulphur

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the three subchapters of the lecture and end with a pharmacy scenario.

Answer key

3.1 · Alcohols: names, classes, properties

Q1 CH3-LO1

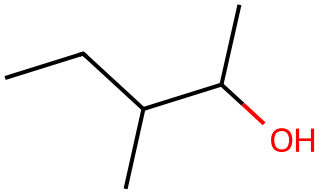
Classify each alcohol.

$CH_3CH_2CH_2OH$ → 1°; $CH_3CH(OH)CH_3$ → 2°; $(CH_3)_3COH$ → 3°; cyclohexanol → 2°

Count the carbons bonded to the C–OH carbon.

Q2 CH3-LO1

Name this alcohol.



A. 3-methyl-2-pentanol

B. 3-methyl-4-pentanol

C. 2-ethyl-1-butanol

D. hexanol

Five-carbon chain numbered from the end nearer –OH: OH on C2, CH_3 on C3 (Ball 14.2).

Q3 CH3-LO1

Arrange in order of increasing water solubility: 1-butanol, methanol, 1-octanol. (Ball 14.E)

A. 1-octanol < 1-butanol < methanol

B. methanol < 1-butanol < 1-octanol

C. 1-butanol < methanol < 1-octanol

D. all equal

Longer chains make alcohols more hydrocarbon-like.

Q4 CH3-LO1

Why is propylene glycol, but not ethylene glycol, used as a drug solvent?

A. Propylene glycol is metabolised to harmless pyruvate; ethylene glycol to toxic oxalate

B. Ethylene glycol has no –OH

C. Propylene glycol is insoluble in water

D. They are the same compound

Ball 14.6.

3.1 · SN1 and SN2

Q5 CH3-LO2

Match each feature to SN1 or SN2.

one step → SN2; goes through a carbocation → SN1; best for 1° haloalkanes → SN2; best for 3° haloalkanes → SN1; rate depends on [R–X] only → SN1; inverts a chiral carbon → SN2

SN2: one step, backside attack, 1°, both species in the rate. SN1: carbocation, 3°, rate on R–X only.

Q6 CH3-LO2

Which haloalkane reacts fastest by SN2?

A. CH_3CH_2Br

B. $(CH_3)_3CBr$

C. $(CH_3)_2CHBr$

D. they are equal

Primary: least crowded for backside attack.

Q7 CH3-LO2

Which is the best leaving group?

A. I^-

B. Cl^-

C. F^-

D. OH^-

$I^- > Br^- > Cl^-$; OH^- is poor.

Q8 CH3-LO2

Name the alcohol formed when $(CH_3)_3CBr$ is warmed with water.

2-methyl-2-propanol

SN1: carbocation $(CH_3)_3C^+$, then water attacks.

Q9 CH3-LO2

Bromoethane is heated with KOH dissolved in ethanol. The main organic product is:

A. ethene

B. ethanol

C. diethyl ether

D. ethanal

Hot ethanolic KOH favours elimination; aqueous OH^- would give ethanol.

3.1 · Dehydration and oxidation

Q10 CH3-LO3

Give the product of each alcohol with acidified $K_2Cr_2O_7$.

2-pentanol → 2-pentanone; cyclohexanol → cyclohexanone; 2-methyl-2-propanol → no reaction; ethanol (excess oxidant) → ethanoic acid

2° → ketone; 3° → none; 1° with excess oxidant → acid (Ball 14.E).

Q11 CH3-LO3

2-Methyl-2-propanol with excess conc. H_2SO_4 at 180 °C gives: (Ball 14.E)

A. 2-methylpropene (isobutylene)

B. 2-methylpropanal

C. di-tert-butyl ether

D. no reaction

Dehydration (elimination of water) to the alkene.

Q12 CH3-LO3

$C_4H_{10}O$ → C_4H_8O is an oxidation. (Ball 14.E)

True / False → **True**

Two H atoms are removed.

Q13 CH3-LO3

Why is methanol far more dangerous to drink than ethanol?

A. It is oxidised to formaldehyde and formic acid

B. It is a thiol

C. It cannot be metabolised

D. It is an alkane

These products damage the optic nerve (Ball 14.4).

3.2 · Ethers and thiols

Q14 CH3-LO4

Name $CH_3OCH_2CH_2CH_3$. (Ball 14.E)

A. methyl propyl ether

B. butanol

C. propyl methanol

D. methoxyethane

Name both groups, then add 'ether'.

Q15 CH3-LO4

Diethyl ether (bp 35 °C) and 1-butanol (bp 117 °C) are isomers. Which statement is true?

A. Both dissolve in water to a similar small extent (about 8 g/100 mL)

B. The ether hydrogen bonds to itself

C. 1-Butanol is insoluble in water

D. The ether boils higher

Both can accept hydrogen bonds from water (FGOB 14.7).

Q16 CH3-LO4

Write the condensed formula of tert-butyl mercaptan, a natural-gas odorant. (FGOB 14.8)

$(CH_3)_3CSH$

A tert-butyl group bonded to –SH.

Q17 CH3-LO4

Mild oxidation of two thiol molecules gives:

A. a disulfide R–S–S–R

B. an ether

C. a sulfate

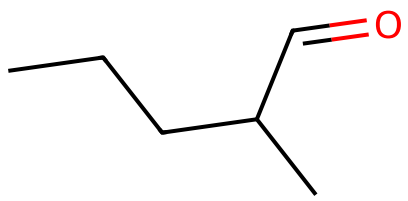
D. an alkene

2 R–SH + [O] → R–S–S–R + H_2O.

3.3 · Aldehydes and ketones

Q18 CH3-LO5

Name this compound.



A. 2-methylpentanal

B. 4-methylpentanal

C. 2-methylpentanone

D. hexanal

The aldehyde carbon is C1, so the methyl is on C2 (FGOB 15.2).

Q19 CH3-LO5

Which has the highest boiling point? (similar molar masses)

A. propanal

B. butane

C. methoxyethane

D. 1-propanol

Hydrogen bonding (alcohol) > dipole (aldehyde) > weak dipole (ether) > dispersion (alkane).

Q20 CH3-LO5

Predict each test result for propanal and propanone.

propanal + Tollens' → silver mirror; propanone + Tollens' → no change; propanal + Benedict's → brick-red precipitate; propanone + Benedict's → no change

Only the aldehyde is oxidised.

Q21 CH3-LO5

Reduction of butanone with H_2/Ni gives:

A. 2-butanol

B. 1-butanol

C. butanal

D. butanoic acid

Ketone → 2° alcohol.

Pharmacy scenario · The diabetic patient

Q22 CH3-LO5

Which compound causes the smell on his breath?

A. acetone (a ketone)

B. ethanol

C. diethyl ether

D. ethanethiol

In uncontrolled diabetes, ketone bodies including acetone build up (FGOB 15.4).

Q23 CH3-LO5

Benedict's reagent detected glucose in urine because glucose behaves like:

A. an aldehyde (a reducing sugar)

B. a ketone that resists oxidation

C. an ether

D. a thiol

Glucose's aldehyde group reduces blue Cu^{2+} to brick-red Cu_2O (Chapter 5).

Q24 CH3-LO5

What should the pharmacist do, and why?

Refer him urgently to a doctor or emergency department: the smell of acetone with nausea in a person with type 1 diabetes who has missed insulin suggests diabetic ketoacidosis, a medical emergency. Do not just sell an anti-nausea product.

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 14 and exercises 14.E, and *Fundamentals of General Organic and Biological Chemistry* 14.4–14.9 and 15.1–15.5, LibreTexts, CC BY-NC-SA 3.0. SN1 and SN2 items, colour tests and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.